

~/portfolio/senior-software-engineer

>_ Senior Software Engineer · ASRS · WMS/WCS

M Luthfi Aditya Putra

I build the software that runs **Automated Storage & Retrieval Systems (ASRS)**: WMS/WCS, hardware integration, APIs and databases, from the first requirement to customer handover.

```
const engineer = { product: 'ASRS', stack: ['Java', 'Spring Boot', 'Oracle'], customers: ['Bank Indonesia', 'Sime Darby'] };
```

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→ luthfi2000.github.io/Portfolio



● open to new roles

📍 Jakarta, ID

DAIFUKU
Automation that Inspires

Current
employer
**PT Daifuku
Indonesia**

3+

Years building enterprise software

2

ASRS customer projects handed over

8→17

Led the Java 8 to 17 migration

E2E

Requirements through go-live and handover

// whoami

About me

cat about.md

Senior Software Engineer with **3+ years** at **PT Daifuku Indonesia**, building software for industrial warehouse automation, especially **ASRS (Automated Storage & Retrieval Systems)** through **WMS** and **WCS**.

Strong in **Java (8 & 17)**, **Spring Boot**, REST/SOAP APIs and relational databases (Oracle, MySQL, PostgreSQL). I own projects across the full lifecycle, from requirements and system design through hardware integration, on-site commissioning and production support, and I've delivered them for **Bank Indonesia Makassar** and **Sime Darby**.

I work comfortably with clients and hardware teams, and I care about dependable, on-time delivery.

ROLE Senior Software Engineer

COMPANY PT Daifuku Indonesia

FOCUS WMS · WCS · ASRS integration

CORE STACK Java · Spring Boot · Oracle

EDUCATION B.Eng. Computer Science,
BINUS

BASED IN Jakarta, Indonesia

What I build: ASRS software

daifuku · intralogistics



At Daifuku I work on the **software layer** that makes automated warehouses run, connecting business logic, inventory systems and physical material-handling equipment.

An **ASRS** stores and retrieves loads from high-density racks automatically, using stacker cranes, conveyors, STVs and robot arms, all coordinated by software:

- **Storage racks:** high-bay structures for dense inventory
- **Stacker cranes & shuttles:** put away and pick loads
- **Conveyors & STVs:** move loads between zones
- **WMS & WCS software:** the logic and real-time control

The software stack I work on

LAYER 01

WMS

Warehouse Management System: inventory, receiving, storage, picking, shipping and operational workflows.

LAYER 02

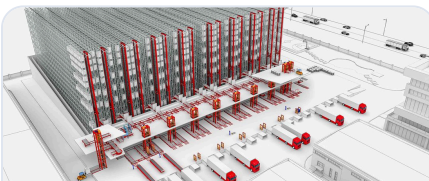
WCS

Warehouse Control System: orchestrates cranes, conveyors, STVs and robot arms in real time.

LAYER 03

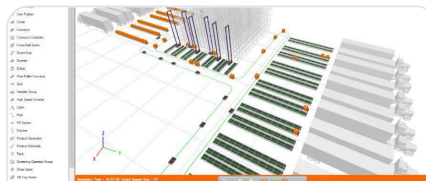
Integration

REST/SOAP APIs, ERP data exchange, proprietary hardware protocols and database design.



ASRS system orchestration

Software decides where goods are stored and retrieved, then coordinates equipment movement safely and efficiently.



Hardware-aware backend

Crane, conveyor, STV and robot arm behavior built into reliable service workflows and communication specs.



Enterprise software delivery

Java/Spring services, relational databases, API contracts, SIT/UAT, commissioning and production support.

My role on ASRS projects

- Design and develop **WMS/WCS features** in native Java for customer warehouse operations.
- Build and maintain **integration points** between software and ASRS equipment (crane, conveyor, STV, robot arm).
- Implement **REST/SOAP APIs** and ERP data exchange for customer-facing systems.
- Model and optimize **Oracle / MySQL / PostgreSQL** structures for performance and maintainability.
- Support the full lifecycle: requirements, design, development, SIT/UAT, commissioning, go-live and handover.
- Troubleshoot production issues across the software, hardware and database layers.

Customer projects delivered

2 of 2 handed over

ASRS WMS/WCS projects I owned end-to-end, through formal handover to the customer.

01 Requirements Business & technical needs	02 Design System & interface specs	03 Development WMS/WCS & APIs	04 Integration Equipment + ERP
05 SIT Testing & RCA	06 UAT Customer acceptance	07 Commissioning On-site readiness	08 Handover Go-live & KT

01 • Banking

✓ Handed over



Bank Indonesia, Makassar

Central bank • Warehouse automation / ASRS software

INDUSTRY	LOCATION
Banking / Central bank	Makassar, Indonesia
DOMAIN	OUTCOME
ASRS • WMS/WCS	Delivered & handed over

- Owned delivery from requirements and design through SIT/UAT, commissioning and handover.
- Developed and adapted native-Java WMS/WCS components for operational workflows.
- Supported ASRS equipment integration and customer-facing API/ERP flows.
- Handled production-readiness troubleshooting and formal knowledge transfer.

Banking

ASRS

WMS/WCS

Handover

02 • Industrial

✓ Handed over



Sime Darby

Industrial / Logistics • Warehouse automation / ASRS software

INDUSTRY	FOCUS
Industrial / Logistics	ASRS software & integration
DOMAIN	OUTCOME
WMS/WCS • material handling	Delivered & handed over

- Involved from day one: requirements, solution design and WMS/WCS backend development.
- Integrated the software with material-handling equipment (crane, conveyor, STV, robot per site).
- Supported environment setup, SIT/UAT, stabilization and go-live readiness.
- Completed the formal handover to the customer.

Industrial

ASRS

Integration

Handover

// impact

Engineering highlights



Java 8 → 17 migration

Led the JDK migration of the WMS/WCS codebase from Java 8 to Java 17.



Security hardening

Resolved vulnerabilities flagged by Nexus IQ and Checkmarx scanning.



Hardware integration

Worked with hardware engineers on end-to-end workflows and communication specs for crane, conveyor, STV and robot arm.



Database performance

Designed and optimized Oracle, MySQL and PostgreSQL structures for performance and maintainability.



Agile delivery

Sprint planning and backlog grooming in Azure DevOps Boards; code reviews and branching in Azure Repos and GitHub.



Estimation & after-sales

Estimated software cost for new inquiries and change requests, and oversaw IT operations and after-sales support.

// stack

Core expertise

01

ASRS software

Software for automated storage and retrieval warehouse systems.

02

WMS & WCS

Inventory logic plus real-time equipment orchestration.

03

Hardware integration

Crane, conveyor, STV and robot arm protocols.

04

Java & Spring Boot

Java 8 & 17, OOP design, JUnit, maintainable services.

05

APIs & ERP

REST/SOAP, ERP data exchange, real-time sync.

06

Database engineering

Oracle, MySQL and PostgreSQL design and tuning.

LANGUAGES

Java

SQL

C#

JavaScript

C

C++

HTML/CSS

FRAMEWORKS & TOOLS

Spring Boot

JUnit

Apache Tomcat

Azure DevOps

Git / GitHub

SSRS

SSIS

EasyRepro

Gherkin

DATABASES

Oracle

MySQL

PostgreSQL

SYSTEMS

WMS

WCS

ASRS integration

REST / SOAP

ERP integration

MS Dynamics CRM

.NET Web Services

Experience

git log --oneline

Senior Software Engineer

Dec 2022 - Present

 **PT Daifuku Indonesia · Jakarta**

- Design, develop and maintain native-Java **WMS and WCS** integrated with ASRS hardware (crane, conveyor, STV, robot arm) over proprietary protocols.
- Integrate customer systems through **REST and SOAP APIs** for ERP data exchange and real-time synchronization.
- Own the full project lifecycle: requirements, design, development, testing, on-site and off-site commissioning, cloud and on-premises server setup, and production debugging.
- Run root cause analysis, SIT and UAT across concurrent projects, and refactor legacy components for stability.
- Led the **Java 8 → 17** migration and fixed vulnerabilities flagged by Nexus IQ and Checkmarx.

Technical Consultant (Contract)

Feb 2022 - Dec 2022

 **PT Graha Technosoft Consulting · Jakarta**

- Built business process automation on **Microsoft Dynamics CRM** with plugins, workflows and JavaScript.
- Designed SSRS reports and C# SSIS packages; tuned SQL queries and stored procedures.
- Built internal and external microservices with **.NET Web Services** for cross-platform data integration.

Technical Consultant (Internship)

Feb 2021 - Feb 2022

 **PT Graha Technosoft Consulting · Jakarta**

- Built a UI automation framework with **EasyRepro and Gherkin** to speed up the support team's QA.
- Developed an HTML reporting dashboard, fixed Dynamics CRM defects and supported go-lives.

Owner & Manager

Circle Place Coffee Shop
2020 - 2021 · Pekanbaru

Promotion Team

BINUS University Learning
Center
2019 - 2020 · Jakarta

Network Technician

PT Sumidhaz Permata
Bunda
2016 - 2017 · Pekanbaru

Teaching Assistant

PT Kumon Indonesia
2015 - 2016 · Pekanbaru

Education

BINUS University (Bina Nusantara)**B.Eng. Computer Science · 2018–2022**

GPA 3.21 / 4.00 · Specialization in Database Systems &
Digital Transformation

SMK Negeri 2 Pekanbaru**Computer & Network Engineering · 2015–2018**

Networking fundamentals and hands-on systems work